

ROBOT DESIGN AND APPLICATIONS



ROBOT TASARIMI VE UYGULAMALARI

Robot Tasarımı ve Uygulamaları Dersi | 2025-2026 Güz Dönemi

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Bölüm: Bilgisayar Mühendisliği

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Agenda

5.1 Introduction

5.2 Kinematic Bicycle Model

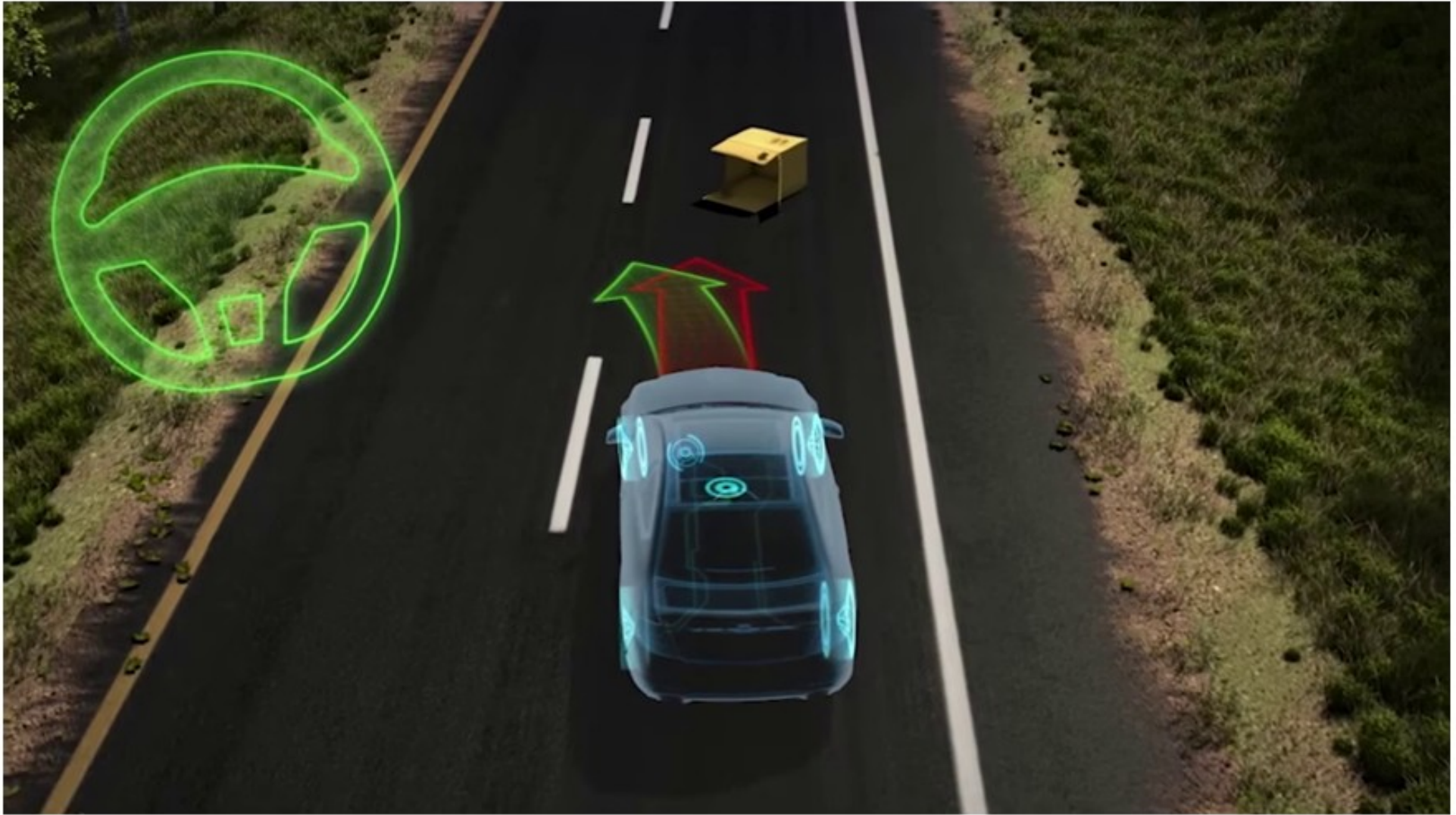
5.3 Tire Models

5.4 Dynamic Bicycle Model

5.1

Introduction

Electronic Stability Program



Knowledge of **vehicle dynamics** enables accurate **vehicle control**

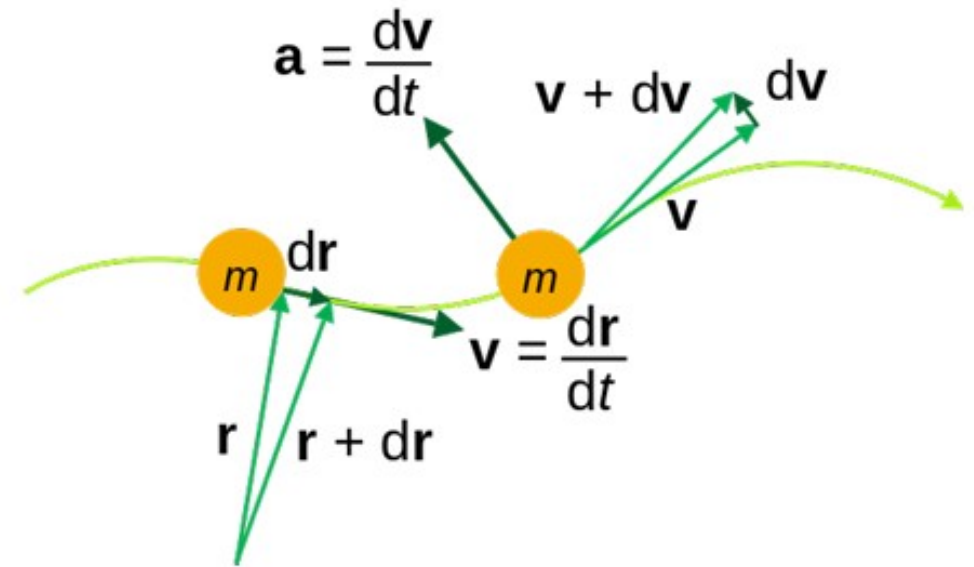
Kinematics vs. Kinetics

Kinematics:

- ▶ Greek origin: “motion”, “moving”
- ▶ Describes motion of points and bodies
- ▶ Considers position, velocity, acceleration, ..
- ▶ Examples: Celestial bodies, particle systems, robotic arm, human skeleton

Kinetics:

- ▶ Describes causes of motion
- ▶ Effects of forces/moments
- ▶ Newton's laws, e.g., $F = ma$



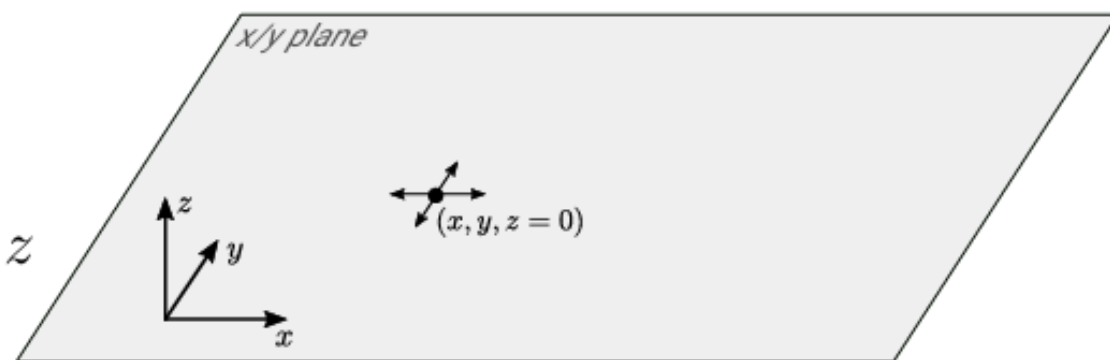
Holonomic Constraints

Holonomic constraints are constraints on the **configuration**:

- ▶ Assume a particle in three dimensions $(x, y, z) \in \mathbb{R}^3$
- ▶ We can constrain the particle to the x/y plane via:

$$z = 0$$

$$\Leftrightarrow f(x, y, z) = 0 \quad \text{with} \quad f(x, y, z) = z$$



- ▶ Constraints of the form $f(x, y, z) = 0$ are called holonomic constraints
- ▶ They constrain the configuration space
- ▶ But the system can move freely in that space
- ▶ Controllable degrees of freedom equal total degrees of freedom (2)

Non-Holonomic Constraints

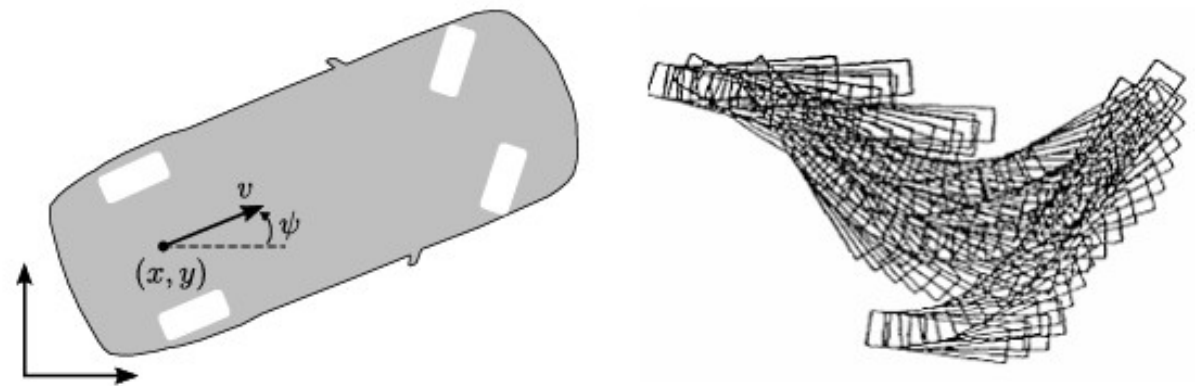
Non-Holonomic constraints are constraints on the **velocity**:

- ▶ Assume a vehicle that is parameterized by $(x, y, \psi) \in \mathbb{R}^2 \times [0, 2\pi]$
- ▶ The 2D vehicle velocity is given by:

$$\dot{x} = v \cos(\psi)$$

$$\dot{y} = v \sin(\psi)$$

$$\Rightarrow \dot{x} \sin(\psi) - \dot{y} \cos(\psi) = 0$$



- ▶ This non-holonomic constraint cannot be expressed in the form $f(x, y, \psi) = 0$
- ▶ The car cannot freely move in any direction (e.g., sideways)
- ▶ It constrains the velocity space, but not the configuration space
- ▶ Controllable degrees of freedom less than total degrees of freedom (2 vs. 3)

Holonomic vs. Non-Holonomic Systems

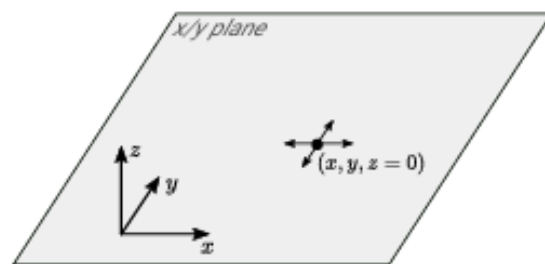
Holonomic Systems

- ▶ Constrain configuration space
- ▶ Can freely move in any direction
- ▶ Controllable degrees of freedom equal to total degrees of freedom
- ▶ Constraints **can** be described by
$$f(x_1, \dots, x_N) = 0$$

Example:

3D Particle

$$z = 0$$



Nonholonomic Systems

- ▶ Constrain velocity space
- ▶ Cannot freely move in any direction
- ▶ Controllable degrees of freedom less than total degrees of freedom
- ▶ Constraints **cannot** be described by
$$f(x_1, \dots, x_N) = 0$$

Example:

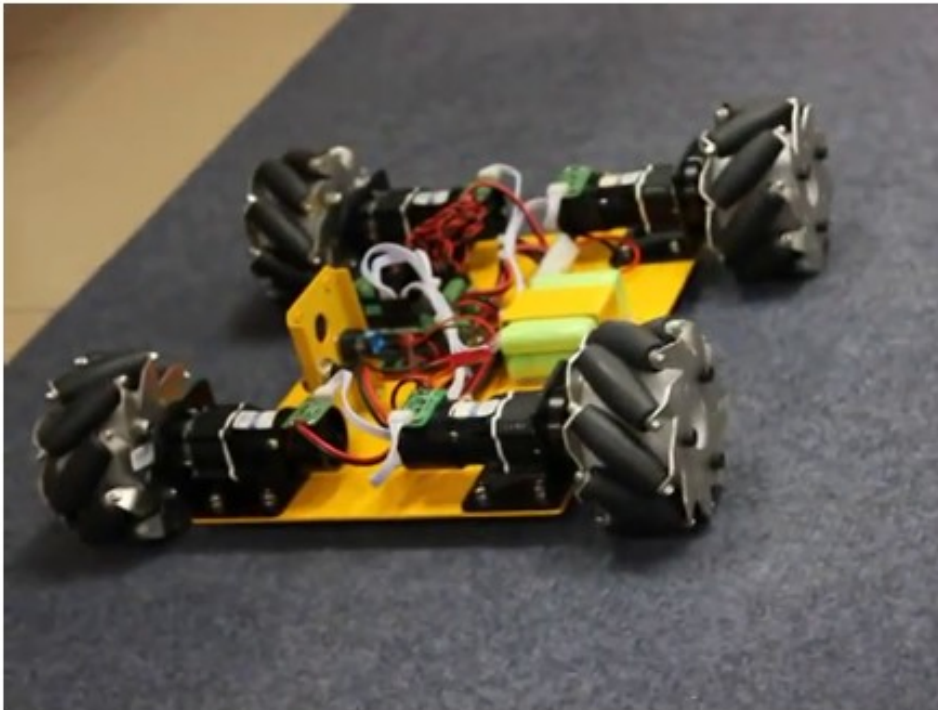
Car

$$\dot{x} \sin(\psi) - \dot{y} \cos(\psi) = 0$$

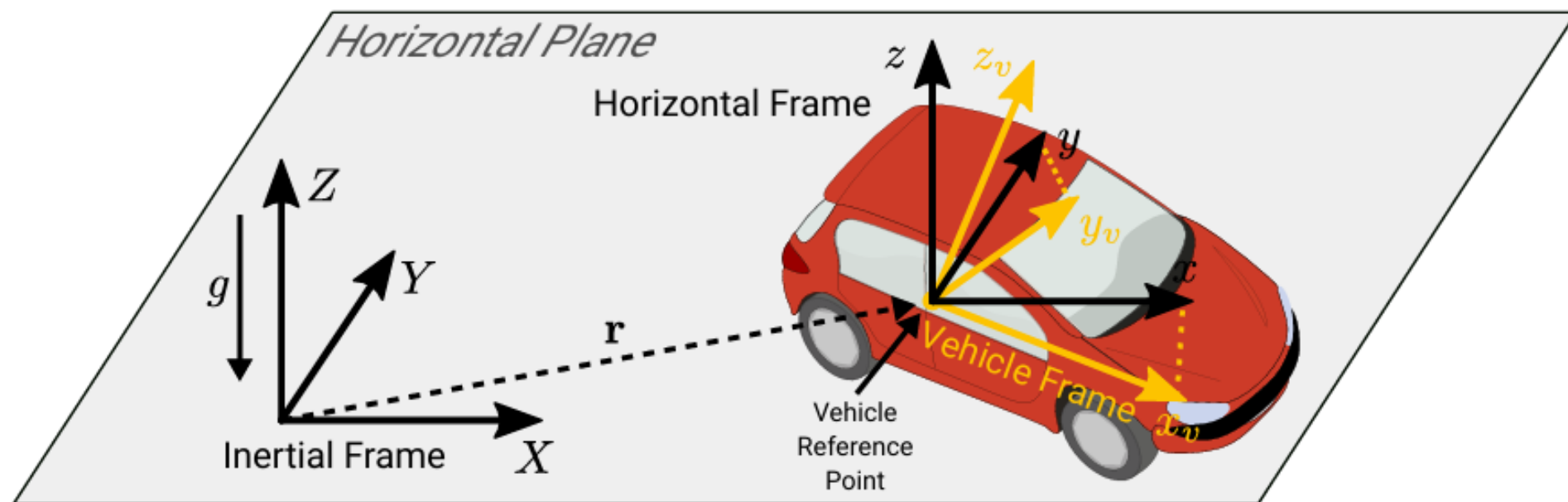


Holonomic vs. Non-Holonomic Systems

- ▶ A robot can be subject to both holonomic and non-holonomic constraints
- ▶ A car (rigid body in 3D) is kept on the ground by 3 holonomic constraints
- ▶ One additional non-holonomic constraint prevents sideways sliding



Coordinate Systems



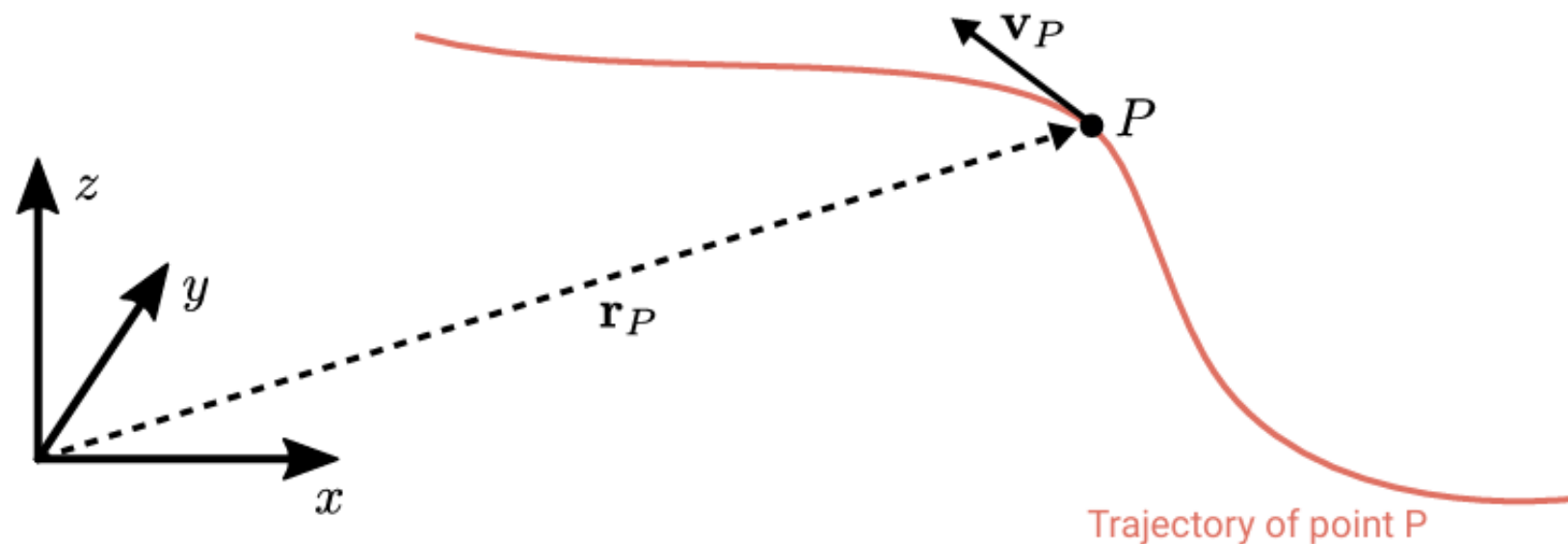
- **Inertial Frame:** Fixed to earth with vertical Z -axis and X/Y horizontal plane
- **Vehicle Frame:** Attached to vehicle at fixed reference point; x_v points towards the front, y_v to the side and z_v to the top of the vehicle (ISO 8855)
- **Horizontal Frame:** Origin at vehicle reference point (like vehicle frame) but x - and y -axes are projections of x_v - and y_v -axes onto the X/Y horizontal plane

Kinematics of a Point

The **position** $\mathbf{r}_P(t) \in \mathbb{R}^3$ of point P at time $t \in \mathbb{R}$ is given by 3 coordinates.

Velocity and **acceleration** are the first and second derivatives of the position $\mathbf{r}_P(t)$.

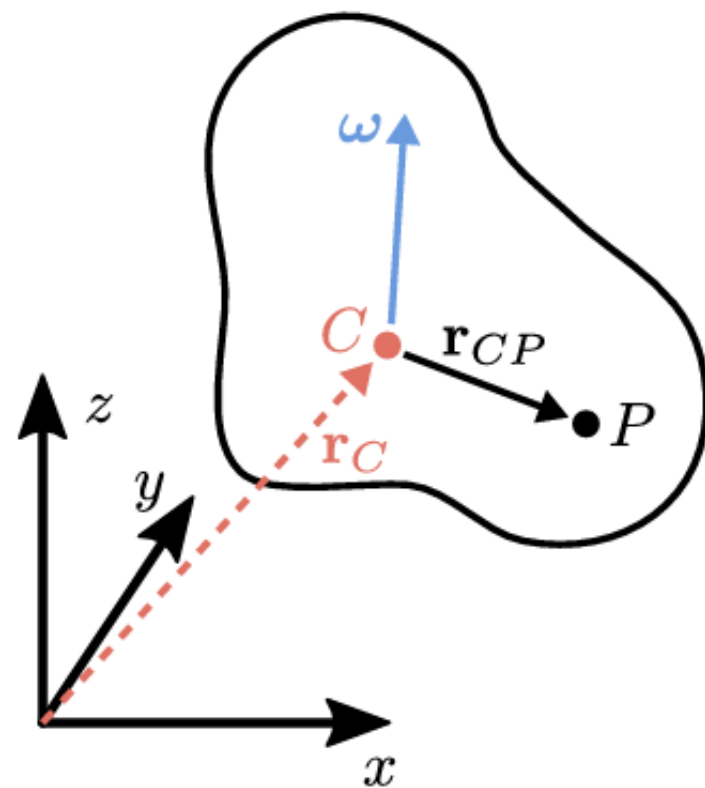
$$\mathbf{r}_P(t) = \begin{pmatrix} x(t) \\ y(t) \\ z(t) \end{pmatrix} \quad \mathbf{v}_P(t) = \dot{\mathbf{r}}_P(t) = \begin{pmatrix} \dot{x}(t) \\ \dot{y}(t) \\ \dot{z}(t) \end{pmatrix} \quad \mathbf{a}_P(t) = \ddot{\mathbf{r}}_P(t) = \begin{pmatrix} \ddot{x}(t) \\ \ddot{y}(t) \\ \ddot{z}(t) \end{pmatrix}$$



Kinematics of a Rigid Body

A **rigid body** refers to a collection of infinitely many infinitesimally small mass points which are rigidly connected, i.e., their relative position remains unchanged over time. It's **motion** can be compactly described by the motion of an (arbitrary) reference point C of the body plus the relative motion of all other points P with respect to C .

- ▶ C : Reference point fixed to rigid body
- ▶ P : Arbitrary point on rigid body
- ▶ ω : Angular velocity of rigid body
- ▶ Position: $\mathbf{r}_P = \mathbf{r}_C + \mathbf{r}_{CP}$
- ▶ Velocity: $\mathbf{v}_P = \mathbf{v}_C + \omega \times \mathbf{r}_{CP}$
- ▶ Due to rigidity, points P can only rotate wrt. C
- ▶ Thus a rigid body has 6 DoF (3 pos., 3 rot.)



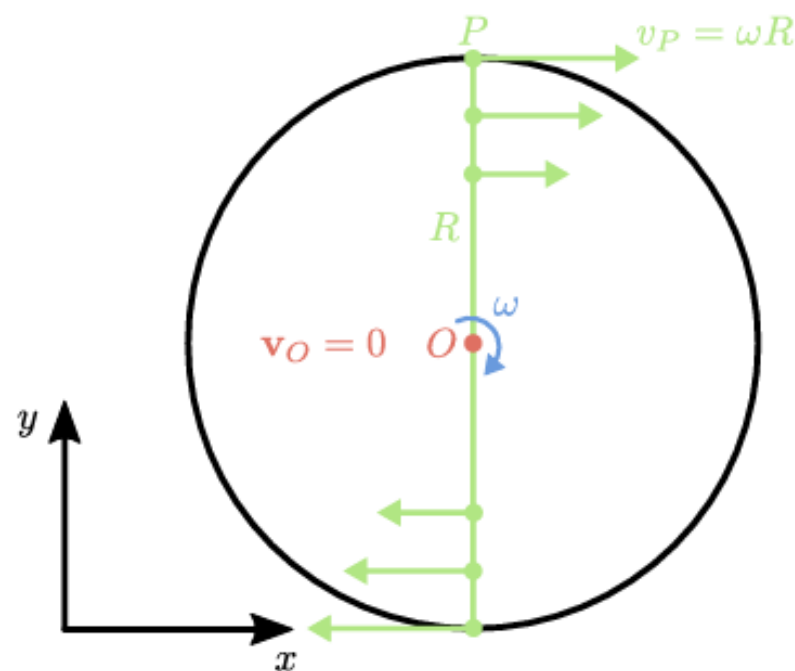
Instantaneous Center of Rotation

At each time instance $t \in \mathbb{R}$, there exists a particular reference point O (called the **instantaneous center of rotation**) for which $\mathbf{v}_O(t) = 0$. Each point P of the rigid body performs a pure rotation about O :

$$\mathbf{v}_P = \mathbf{v}_O + \boldsymbol{\omega} \times \mathbf{r}_{OP} = \boldsymbol{\omega} \times \mathbf{r}_{OP}$$

Example 1: Turning Wheel

- ▶ Wheel is completely lifted off the ground
- ▶ Wheel does not move in x or y direction
- ▶ Ang. vel. vector $\boldsymbol{\omega}$ points into x/y plane
- ▶ Velocity of point P : $v_P = \omega R$ with radius R



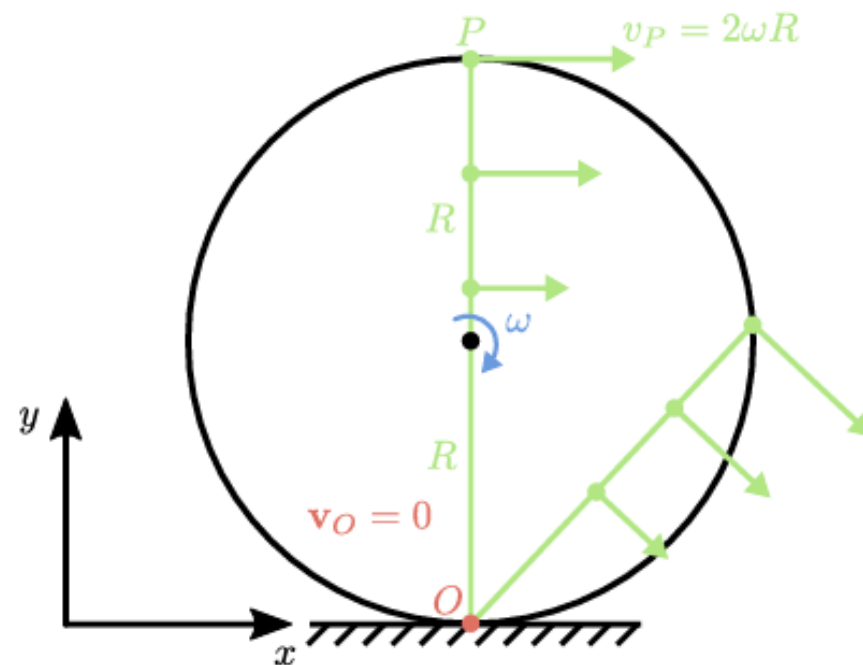
Instantaneous Center of Rotation

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$$\mathbf{v}_P = \mathbf{v}_O + \boldsymbol{\omega} \times \mathbf{r}_{OP} = \boldsymbol{\omega} \times \mathbf{r}_{OP}$$

Example 2: Rolling Wheel

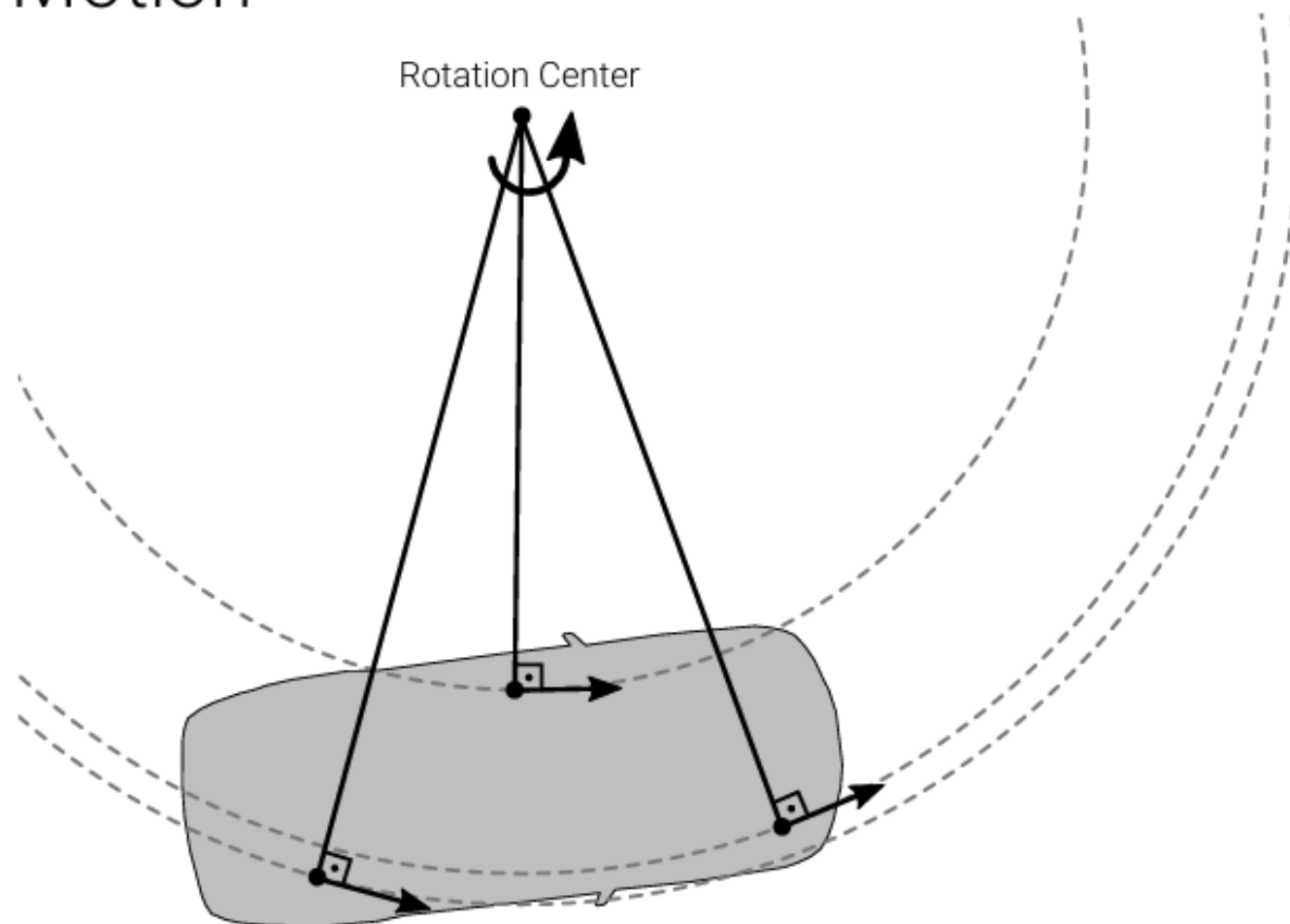
- ▶ Wheel is rolling on the ground without slip
- ▶ Ground is fixed in x/y plane
- ▶ Ang. vel. vector $\boldsymbol{\omega}$ points into x/y plane
- ▶ Velocity of point P: $v_P = 2\omega R$ with radius R



5.2

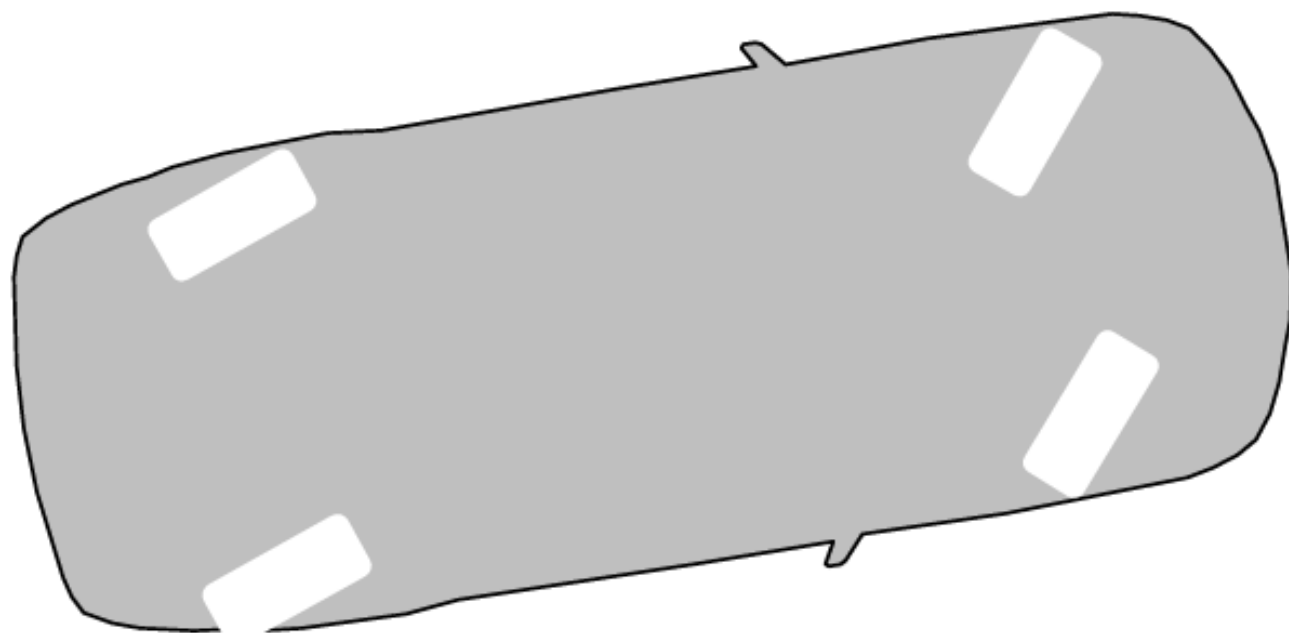
Kinematic Bicycle Model

Rigid Body Motion



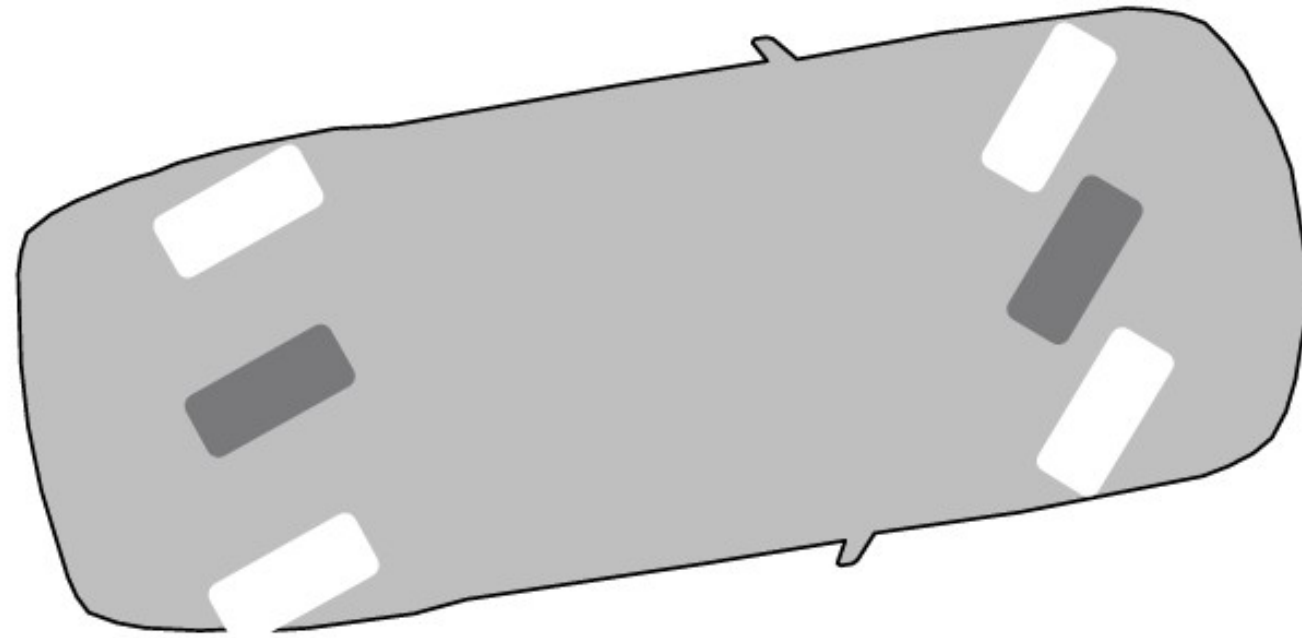
- Different points on the rigid body move along different circular trajectories

Kinematic Bicycle Model



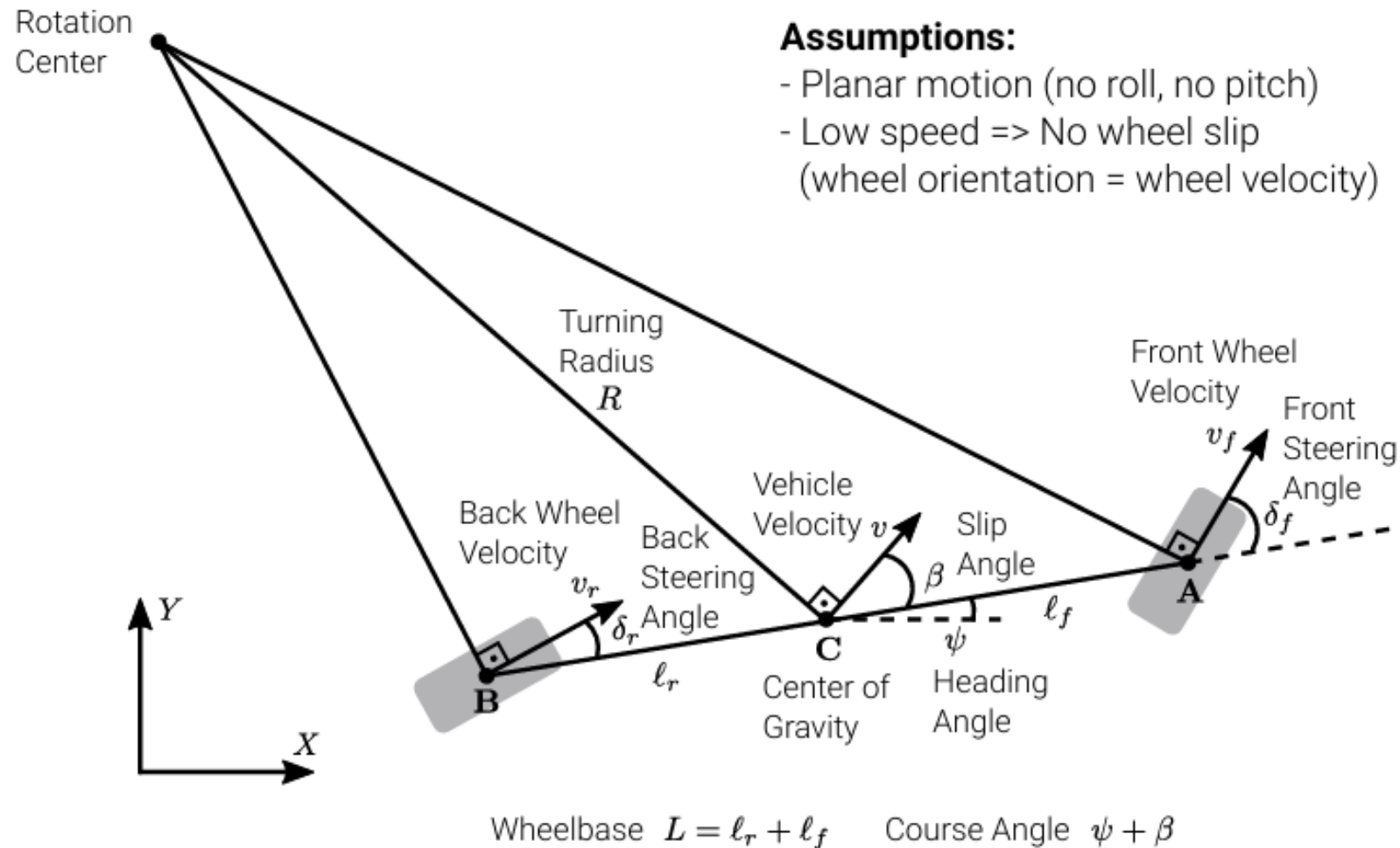
- The **kinematic bicycle model** approximates the 4 wheels with 2 imaginary wheels

Kinematic Bicycle Model



- The **kinematic bicycle model** approximates the 4 wheels with 2 imaginary wheels

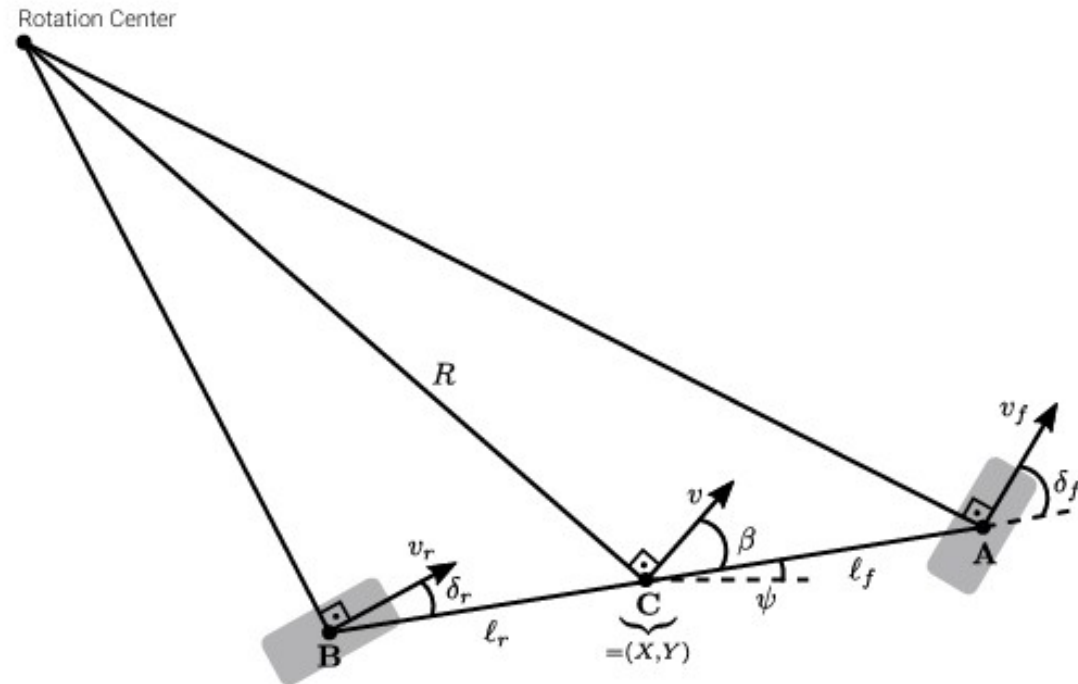
Kinematic Bicycle Model



- The **kinematic bicycle model** approximates the 4 wheels with 2 imaginary wheels

Kinematic Bicycle Model

Model



Motion Equations

$$\dot{X} = v \cos(\psi + \beta)$$

$$\dot{Y} = v \sin(\psi + \beta)$$

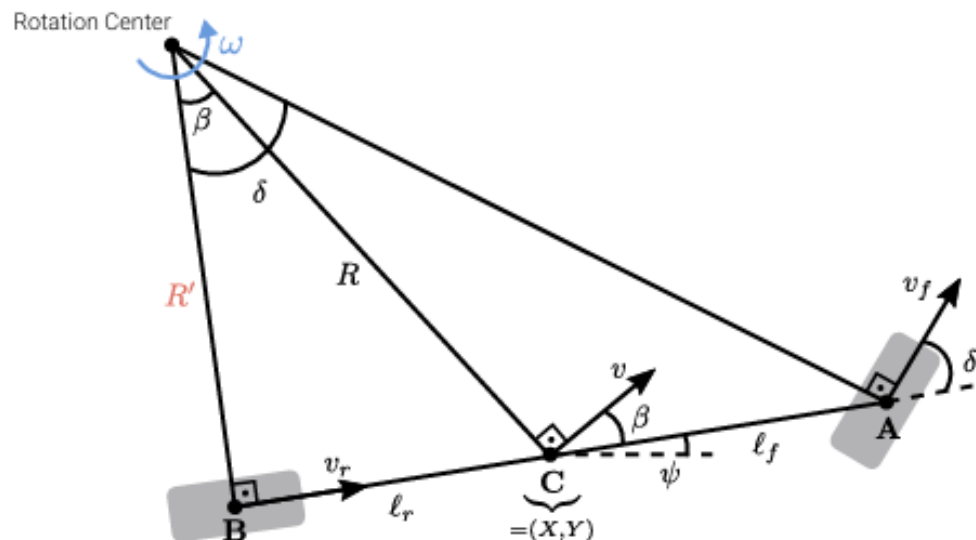
$$\dot{\psi} = \frac{v \cos(\beta)}{\ell_f + \ell_r} (\tan(\delta_f) - \tan(\delta_r))$$

$$\beta = \tan^{-1} \left(\frac{\ell_f \tan(\delta_r) + \ell_r \tan(\delta_f)}{\ell_f + \ell_r} \right)$$

(proof as exercise)

Kinematic Bicycle Model

Model



Motion Equations

$$\dot{X} = v \cos(\psi + \beta)$$

$$\dot{Y} = v \sin(\psi + \beta)$$

$$\dot{\psi} = \frac{v \cos(\beta)}{l_f + l_r} \tan(\delta)$$

$$\beta = \tan^{-1} \left(\frac{l_r \tan(\delta)}{l_f + l_r} \right)$$

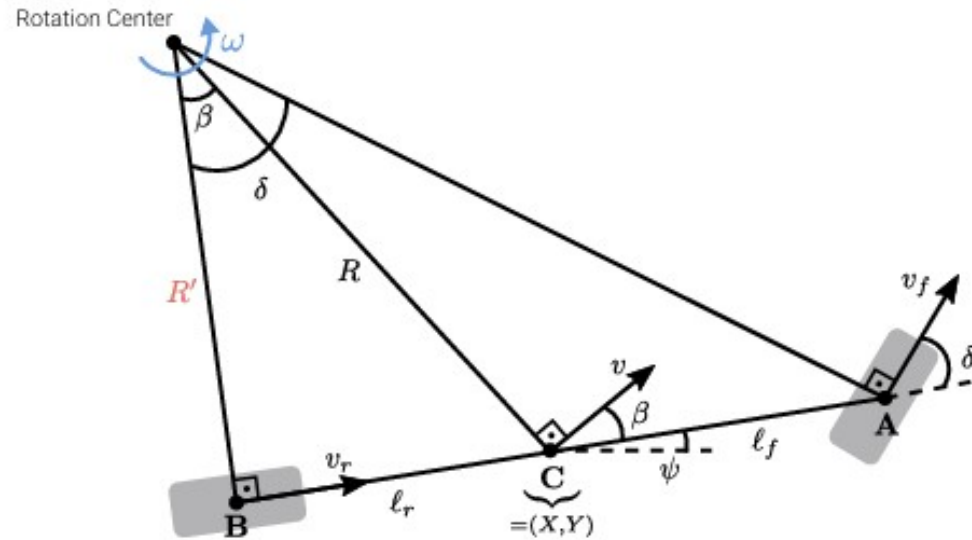
(only front steering)

$$\tan \delta = \frac{l_f + l_r}{R'} \Rightarrow \frac{1}{R'} = \frac{\tan \delta}{l_f + l_r} \Rightarrow \tan \beta = \frac{l_r}{R'} = \frac{l_r \tan \delta}{l_f + l_r}$$

$$\cos \beta = \frac{R'}{R} \Rightarrow \frac{1}{R} = \frac{\cos \beta}{R'} \Rightarrow \dot{\psi} = \omega = \frac{v}{R} = \frac{v \cos(\beta)}{R'} = \frac{v \cos(\beta)}{l_f + l_r} \tan(\delta)$$

Kinematic Bicycle Model

Model



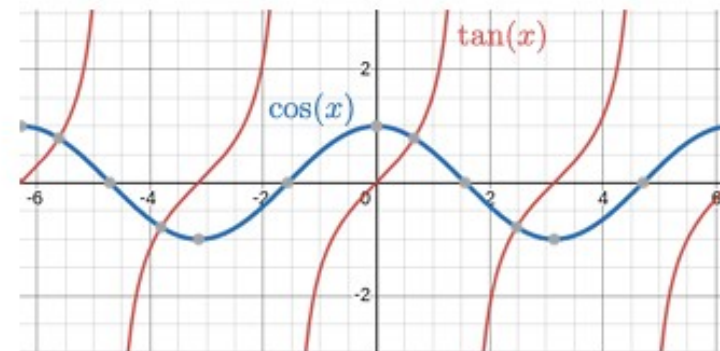
Motion Equations

$$\dot{X} = v \cos(\psi)$$

$$\dot{Y} = v \sin(\psi)$$

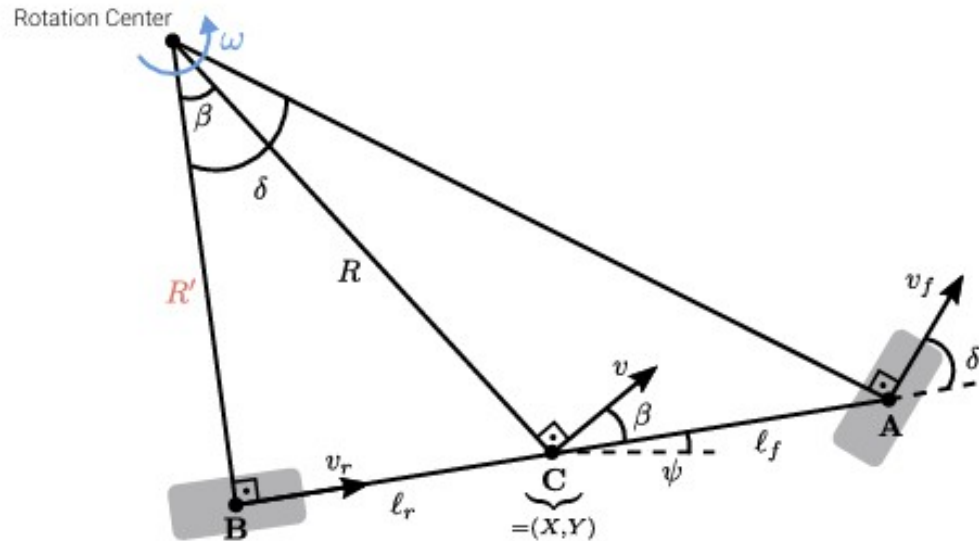
$$\dot{\psi} = \frac{v\delta}{\ell_f + \ell_r}$$

(assuming β and δ are very small)



Kinematic Bicycle Model

Model



Motion Equations

$$X_{t+1} = X_t + v \cos(\psi_t) \Delta t$$

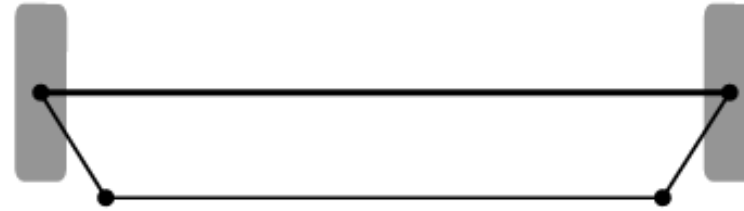
$$Y_{t+1} = Y_t + v \sin(\psi_t) \Delta t$$

$$\psi_{t+1} = \psi_t + \frac{v \delta}{\ell_f + \ell_r} \Delta t$$

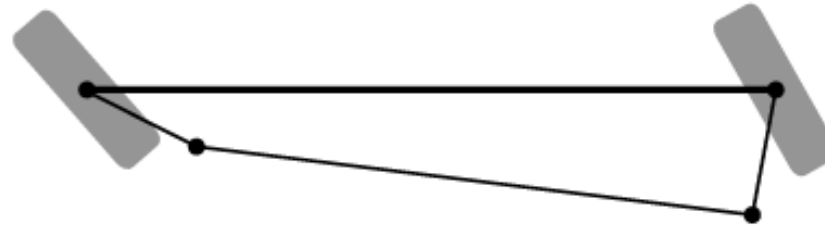
(time discretized model)

Ackermann Steering Geometry

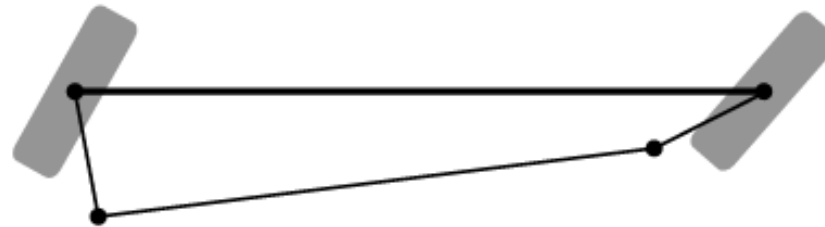
Trapezoidal Geometry



Left Turn



Right Turn

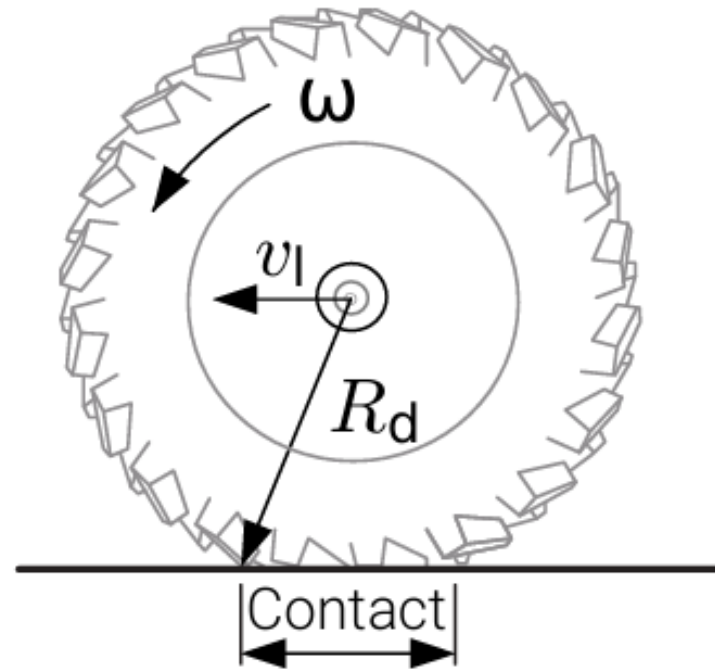


- In practice, this setup can be realized using a trapezoidal tie rod arrangement

5.3

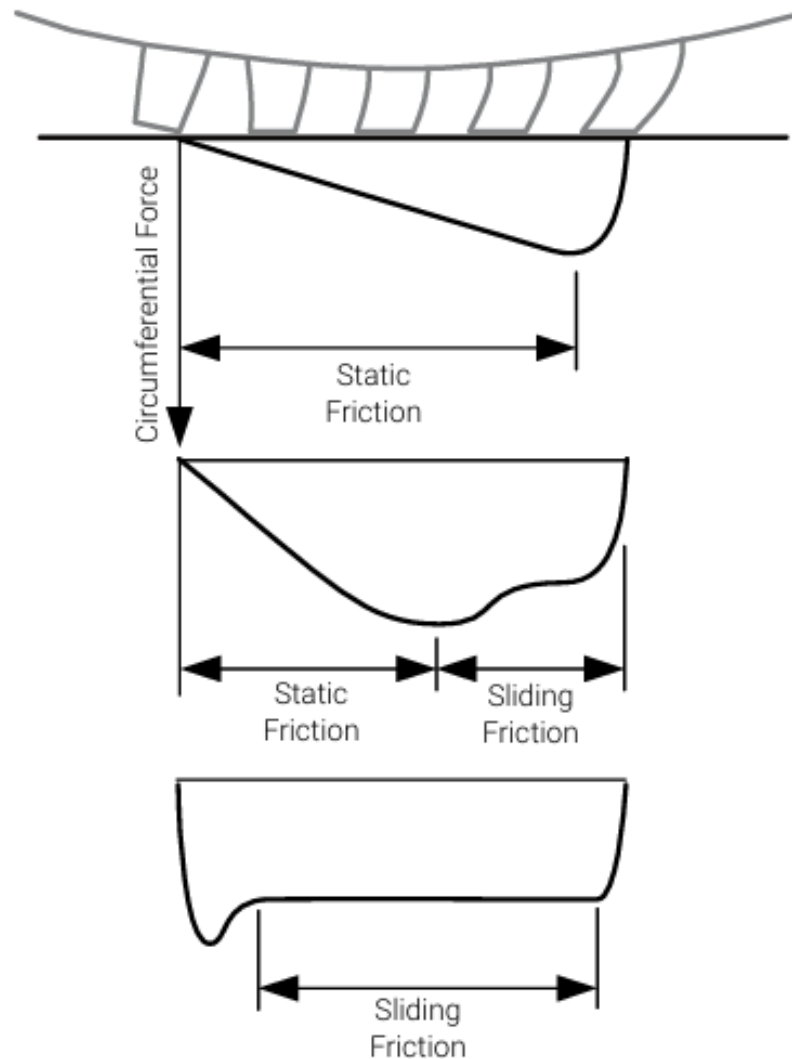
Tire Models

Tire Models



- ▶ Tire models describe the lateral and longitudinal forces at the tires
- ▶ There exist many different tire models at various levels of complexity
- ▶ For a simple qualitative description we consider the **tread block model**
- ▶ **Question:** Why do tires “slip”?

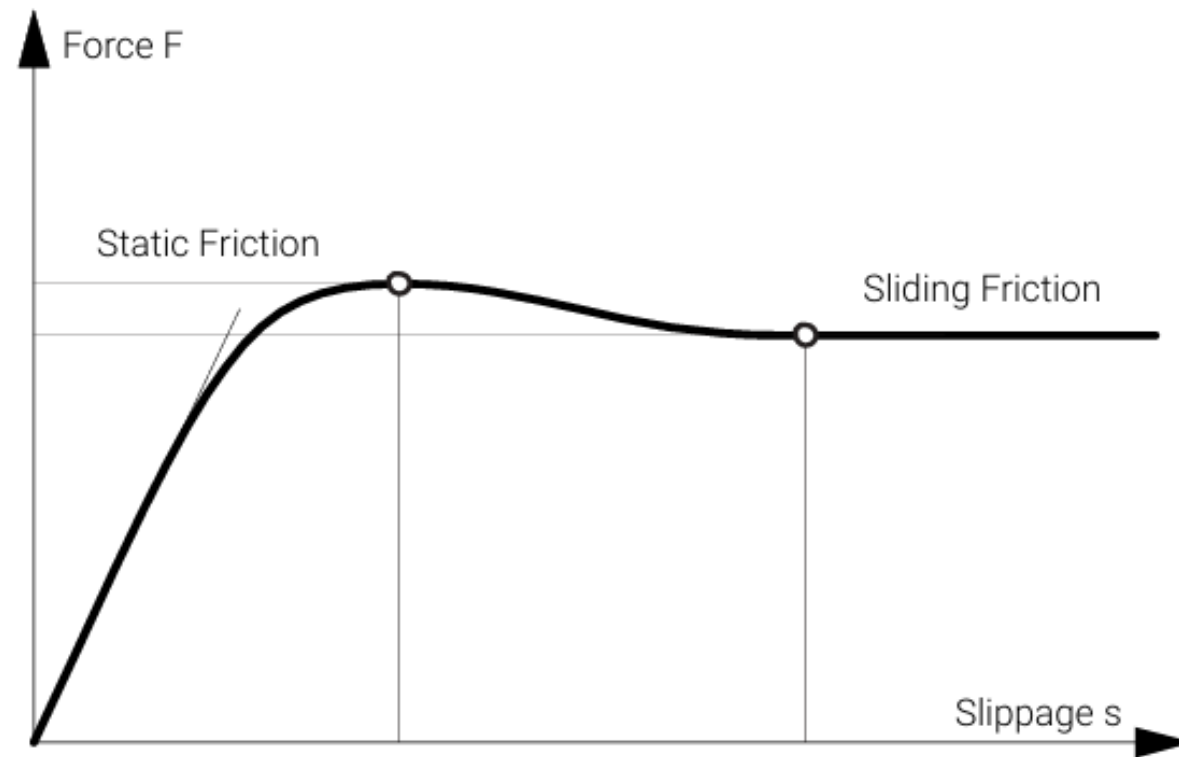
Tread Block Model



Longitudinal Force:

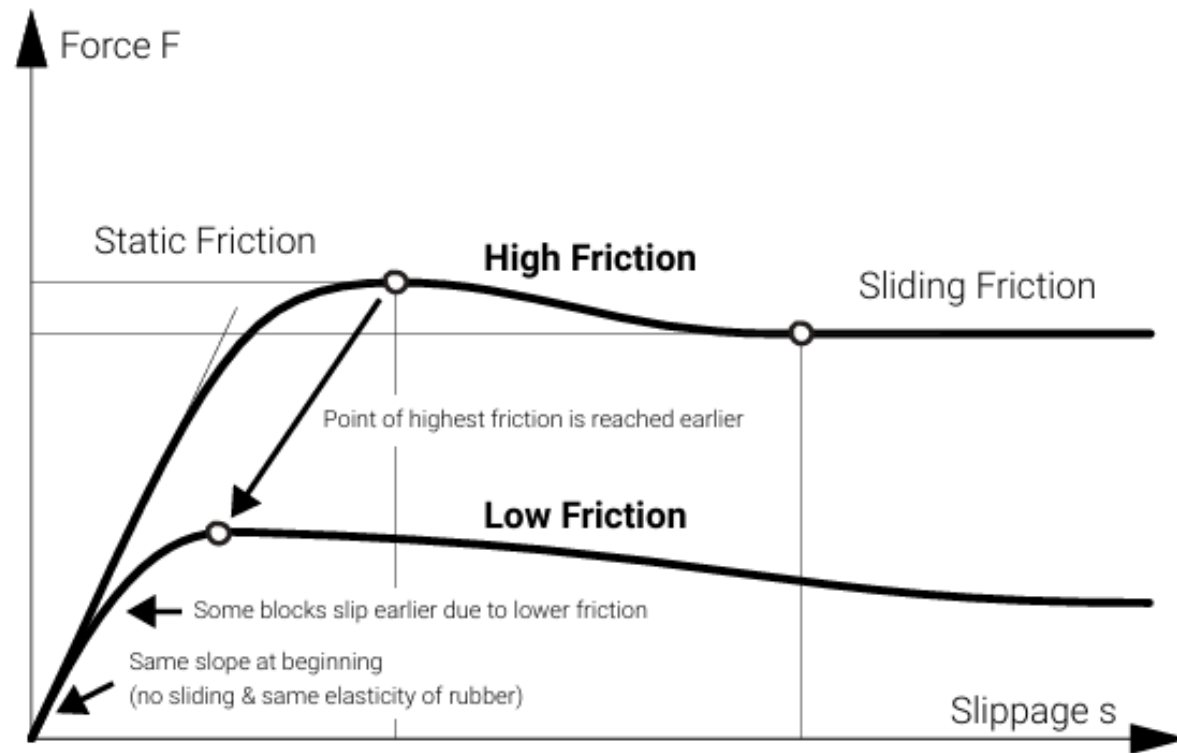
- ▶ As soon as the wheel is driven externally, the **tire tread blocks** start deforming and slipping
- ▶ The tire tread blocks adhere to the ground, **deform** and **slip** when loosing contact
- ▶ When the driving force increases and static friction is exceeded the **blocks slip earlier**
- ▶ As **sliding friction** is smaller than **static friction**, this decreases the transmitted driving force
- ▶ If the tire tread blocks start sliding at the beginning, only **sliding friction** can be applied

Tread Block Model



- ▶ **Slippage:** Difference between surface speed of the wheel and vehicle speed
- ▶ The force F grows **linearly** with the slippage s in the beginning (linear deform.)
- ▶ Large slippage s leads to a **reduction** of F (sliding friction $<$ static friction)

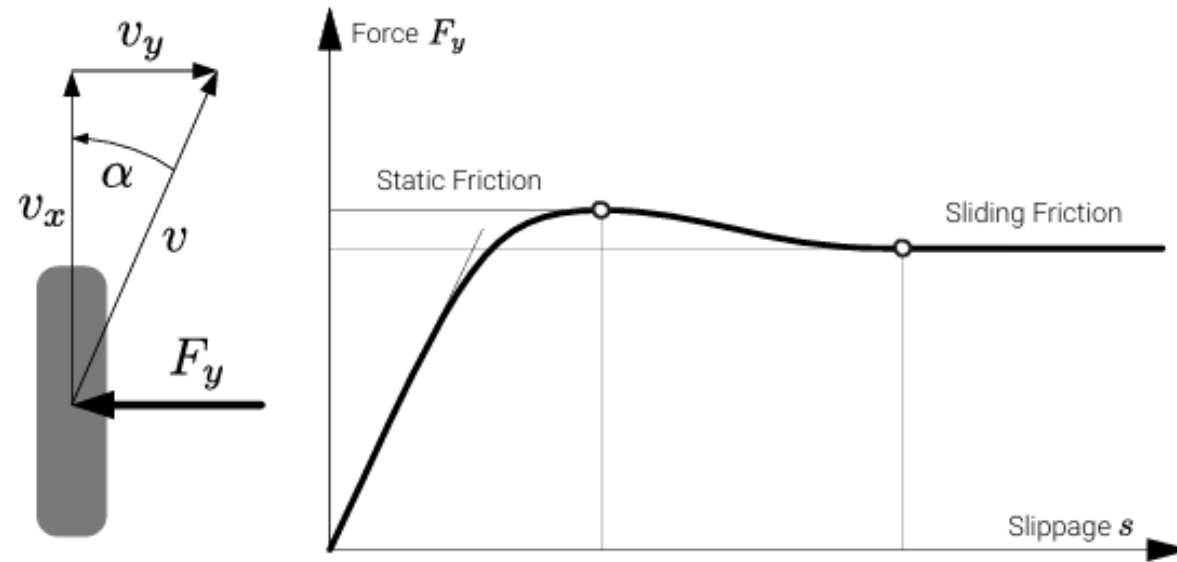
Tread Block Model



How does the force curve $F(s)$ change for **slippery terrain** (low friction)?

- ▶ Start of the curve doesn't change as the elasticity of the blocks doesn't change
- ▶ However, the **maximum reduces** due to the decreased static friction, i.e., the tread blocks start sliding earlier due to a decrease in friction

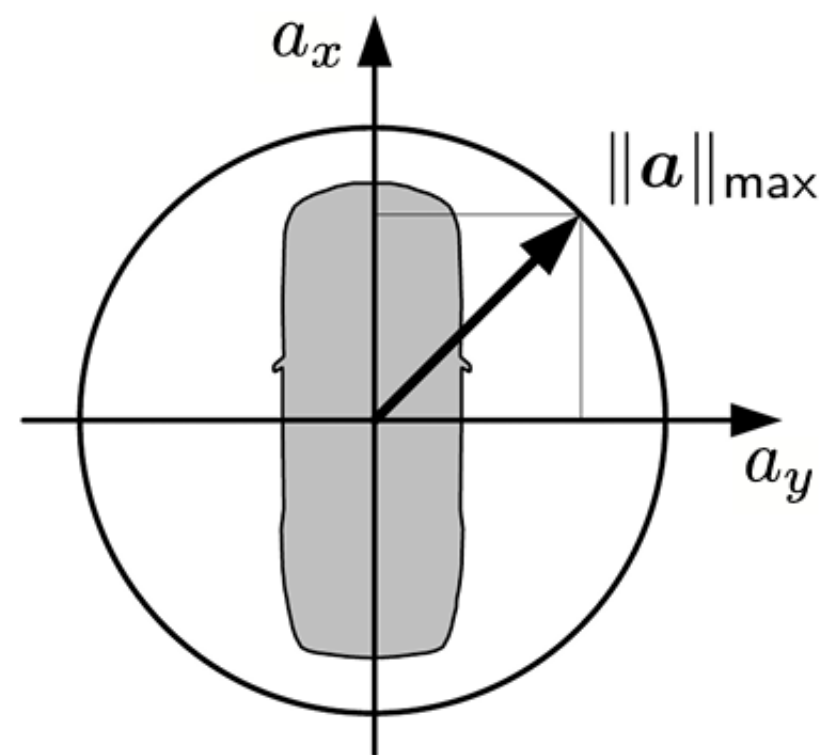
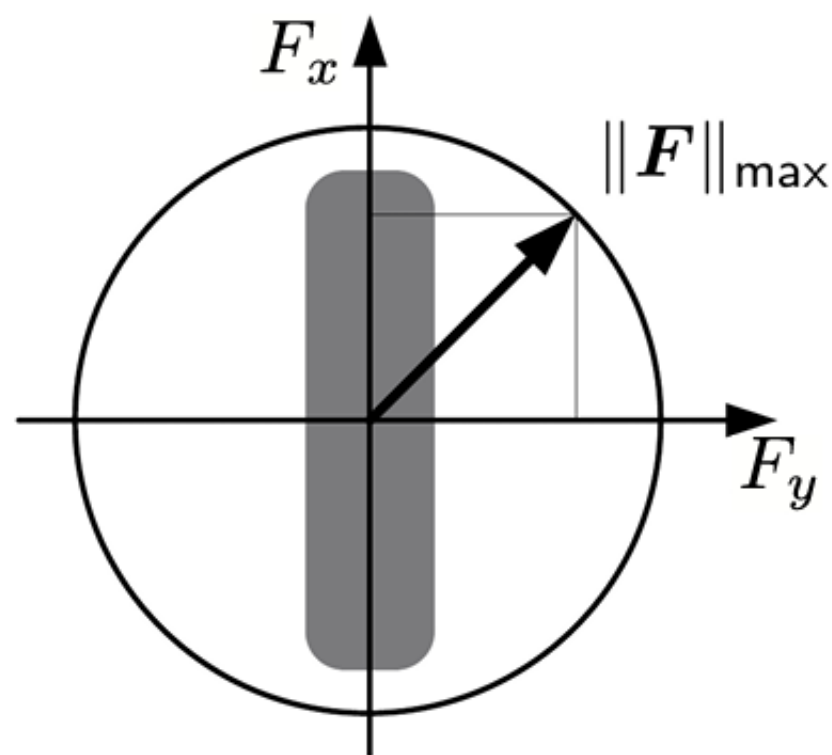
Tread Block Model



Lateral Force: (the figure above shows a top view onto a wheel)

- ▶ Lateral force F_y analogous to longitudinal force but blocks move laterally now
- ▶ Lateral force for small s and α given by: $F_y = c s = c \tan(\alpha) \approx c \alpha$
- ▶ v = wheel velocity, v_x = longitudinal vel., v_y = lateral vel., c = cornering stiffness

Circle of Forces



Circle of Forces:

- ▶ Lateral F_y and longitudinal F_x force cannot exceed max. friction force $\|F\|_{\max}$
- ▶ More long. force implies less lat. force; max. acceleration only for straight driving
- ▶ Allows to make statements about maximal possible vehicle accelerations

5.4

Dynamic Bicycle Model

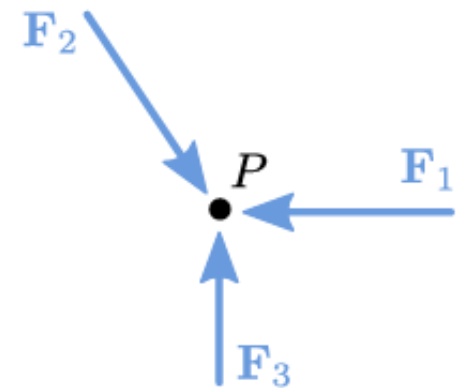
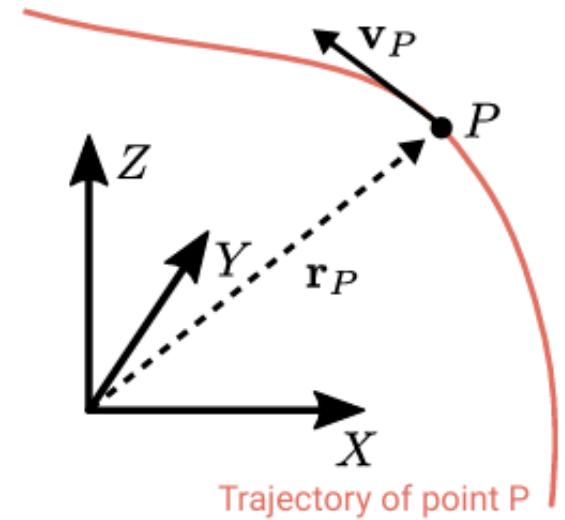
Dynamics of a Rigid Body

Translatory Motion of a Point:

- ▶ Consider **point** P with mass m in $\mathbb{R}^3$
- ▶ Let $\mathbf{r}_P(t) \in \mathbb{R}^3$ be its **position** in an inertial reference frame
- ▶ Let $\mathbf{v}_P(t)$ denote its **velocity** and $\mathbf{a}_P(t)$ its **acceleration**
- ▶ The **linear momentum** of P is defined as $\mathbf{p}_P(t) = m\mathbf{v}_P(t)$
- ▶ By **Newton's second law** we have

$$\frac{d}{dt}\mathbf{p}_P(t) = m\mathbf{a}_P(t) = \mathbf{F}_{net}(t) = \sum_i \mathbf{F}_i(t)$$

where $\mathbf{F}_i(t)$ represent all forces acting on the point mass P



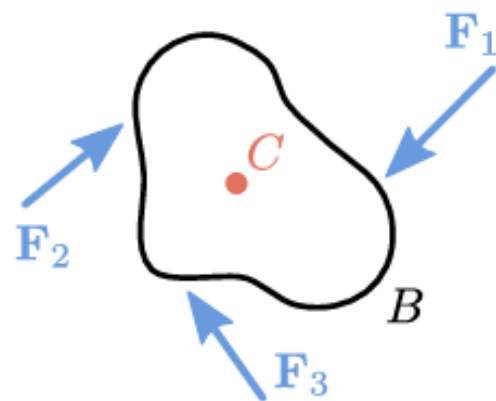
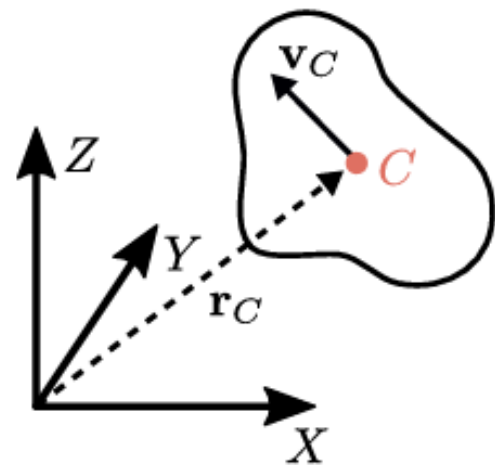
Dynamics of a Rigid Body

Translatory Motion of a Rigid Body:

- ▶ Consider a **rigid body** B with mass m in $\mathbb{R}^3$
- ▶ Let $\mathbf{r}_C(t) \in \mathbb{R}^3$ be the **position** of its **center of gravity** C
- ▶ Let $\mathbf{v}_C(t)$ denote its **velocity** and $\mathbf{a}_C(t)$ its **acceleration**
- ▶ The **linear momentum** of B is defined as $\mathbf{p}_B(t) = m\mathbf{v}_C(t)$
- ▶ The **center of gravity** of a rigid body **behaves like a point mass** with mass m and as if all forces act on that point

$$\frac{d}{dt}\mathbf{p}_B(t) = m\mathbf{a}_C(t) = \mathbf{F}_{net}(t) = \sum_i \mathbf{F}_i(t)$$

where $\mathbf{F}_i(t)$ represent all forces acting on the rigid body B



Dynamics of a Rigid Body

Rotatory Motion of a Rigid Body:

- For the **rotatory motion**, also the geometric shape of B and the spatial distribution of its mass is important
- Let $\rho(x, y, z)$ be the **body's density function**:

$$m = \int_B \rho(x, y, z) dx dy dz = \int_B dm$$

- The **inertia tensor** of B is defined as

$$\Theta = \begin{bmatrix} I_x & I_{xy} & I_{xz} \\ I_{yx} & I_y & I_{yz} \\ I_{zx} & I_{zy} & I_z \end{bmatrix}$$

$$I_x = \int_B (y^2 + z^2) dm$$

$$I_y = \int_B (x^2 + z^2) dm$$

$$I_z = \int_B (x^2 + y^2) dm$$

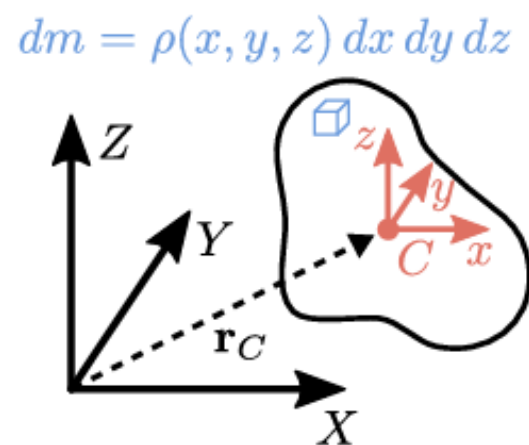
moments of inertia

$$I_{xy} = I_{yx} = - \int_B xy dm$$

$$I_{xz} = I_{zx} = - \int_B xz dm$$

$$I_{yz} = I_{zy} = - \int_B yz dm$$

moments of deviation



Dynamics of a Rigid Body

Rotatory Motion of a Rigid Body:

- ▶ Let ω be the vector of **angular velocities**:

$$\omega = (\omega_x \ \omega_y \ \omega_z)^T$$

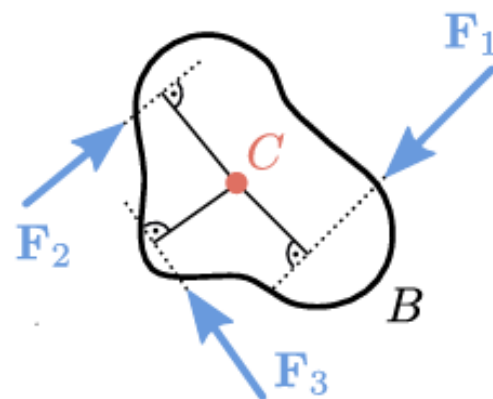
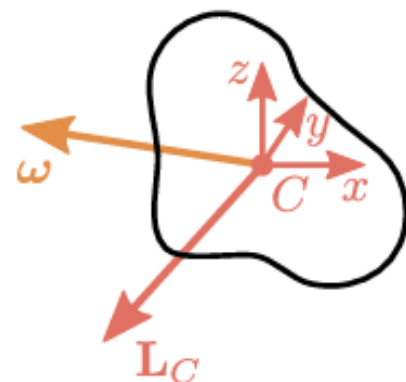
- ▶ The **angular momentum** $\mathbf{L}_C$ of the rigid body B is given by

$$\mathbf{L}_C = \mathbf{\Theta} \omega$$

- ▶ By the **angular momentum principle**

$$\frac{d}{dt} \mathbf{L}_C(t) = \mathbf{\Theta} \dot{\omega} = \mathbf{M}_{net}(t) = \sum_i \mathbf{M}_i(t)$$

where $\mathbf{M}_i(t)$ are the moments of all forces acting on B with respect to the center of gravity C .



Dynamics of a Rigid Body

Rotatory Motion of a Rigid Body with Canonical Coordinates:

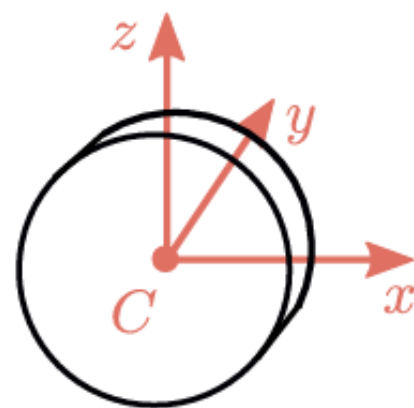
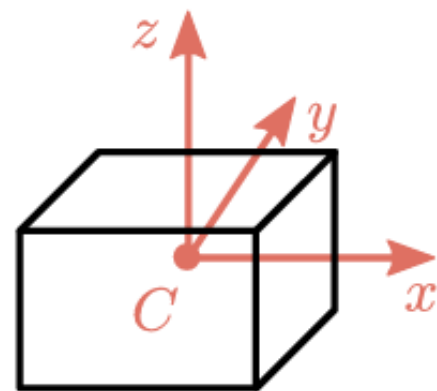
- If the body frame is chosen as a principal axis system for the rigid body (symmetry axes), the inertia tensor is diagonal:

$$\Theta = \begin{bmatrix} I_x & 0 & 0 \\ 0 & I_y & 0 \\ 0 & 0 & I_z \end{bmatrix}$$

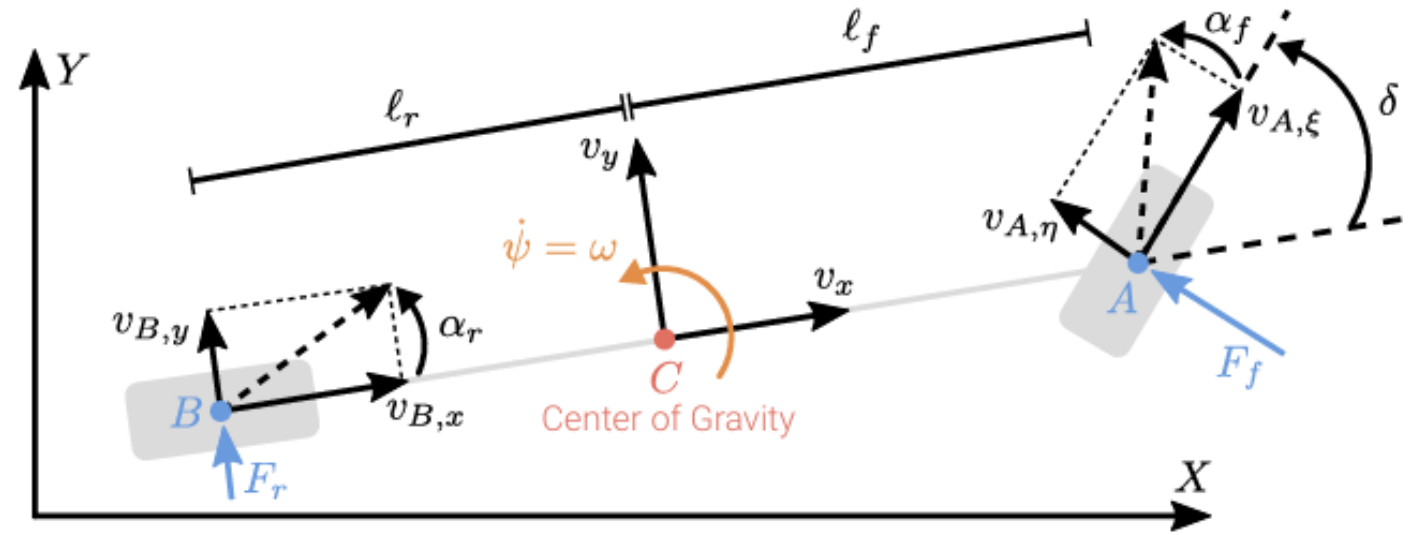
- For the planar motion of a rigid body in the x/y-plane:

$$\omega_x = \omega_y = 0 \quad \text{and} \quad M_x = M_y = 0$$

- Hence the angular momentum becomes $L_z = I_z \omega_z(t)$
and the angular momentum principle yields $I_z \dot{\omega}_z = \sum_i M_i$



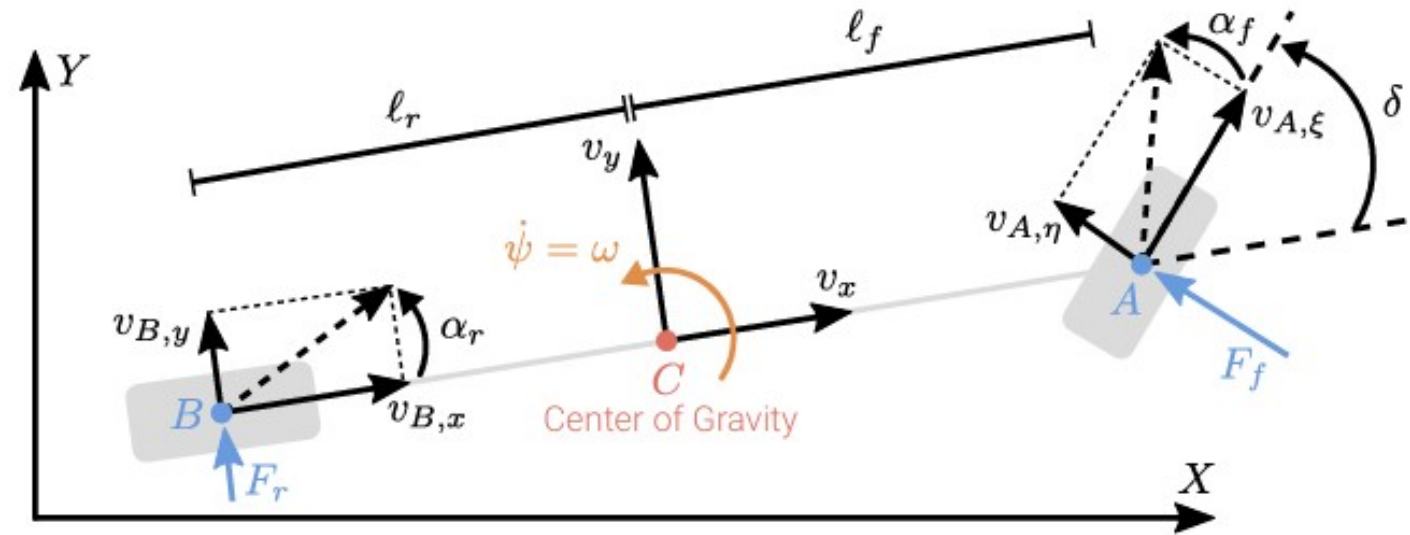
Dynamic Bicycle Model



Assumptions:

- ▶ The vehicle's motion is restricted to the X/Y plane
- ▶ The vehicle is considered as a rigid body
- ▶ Only lateral tire forces, generated by a linear tire model
- ▶ Small steering angle δ : $\sin \delta \approx \delta$ $\tan \delta \approx \delta$ $\cos \delta \approx 1$
- ▶ Constant longitudinal velocity v_x

Dynamic Bicycle Model



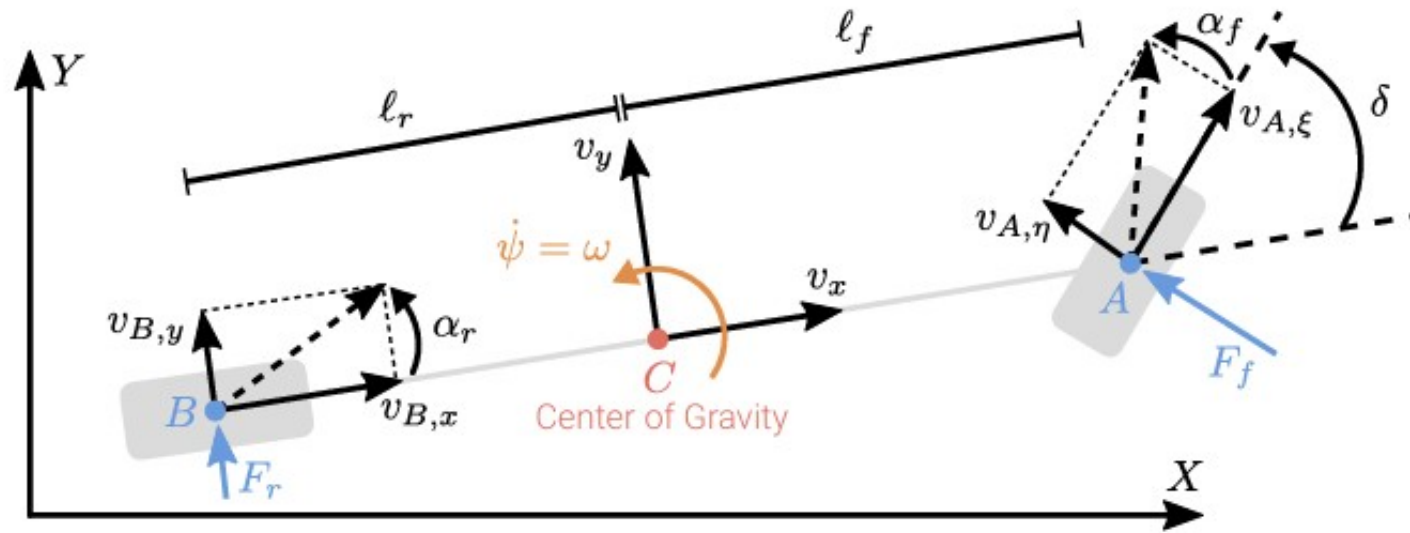
Lateral Dynamics:

$$ma_y = \sum_i F_{y,i} = F_r + F_f \cos \delta \approx F_r + F_f$$

$$a_y = \dot{v}_y + \omega v_x \quad (\omega v_x = \text{centripetal acc.})$$

$$\Rightarrow m(\dot{v}_y + \omega v_x) = F_r + F_f$$

Dynamic Bicycle Model

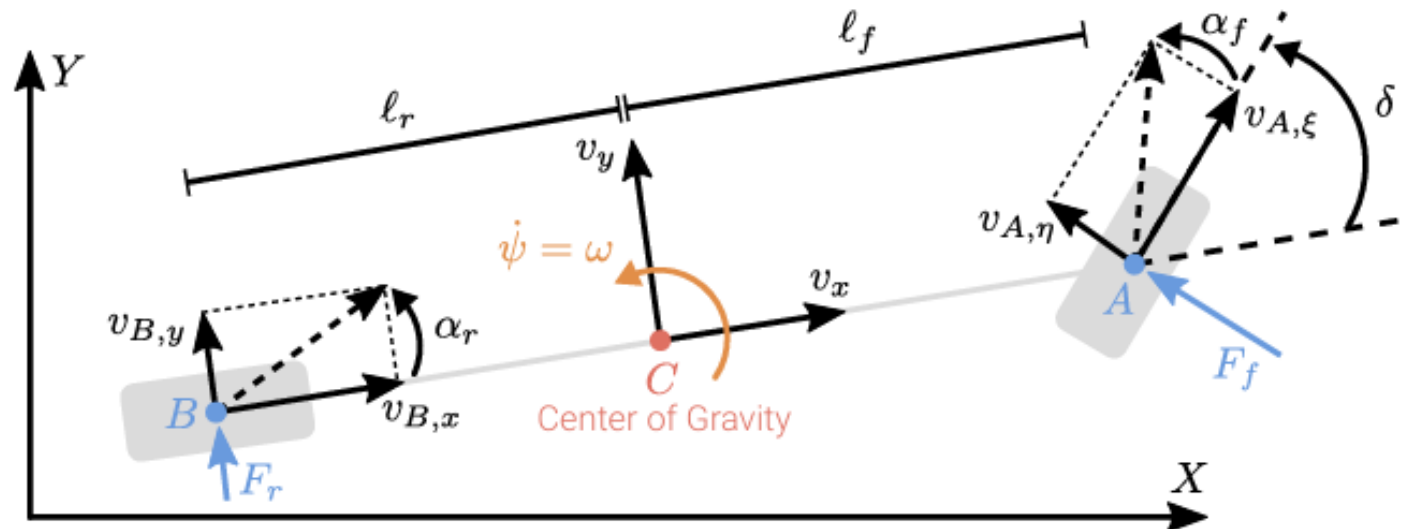


Yaw Dynamics:

$$I_z \dot{\omega} = \sum_i M_i = -l_r F_r + l_f F_f \underbrace{\cos \delta}_{\approx 1}$$

$$\Rightarrow I_z \dot{\omega} = -l_r F_r + l_f F_f$$

Dynamic Bicycle Model



Tire Forces:

$$F_r = -c_r \alpha_r \approx -c_r \tan(\alpha_r) = -c_r \frac{v_{B,y}}{v_{B,x}}$$

$$F_f = -c_f \alpha_f \approx -c_f \tan(\alpha_f) = -c_f \frac{v_{A,\eta}}{v_{A,\xi}}$$

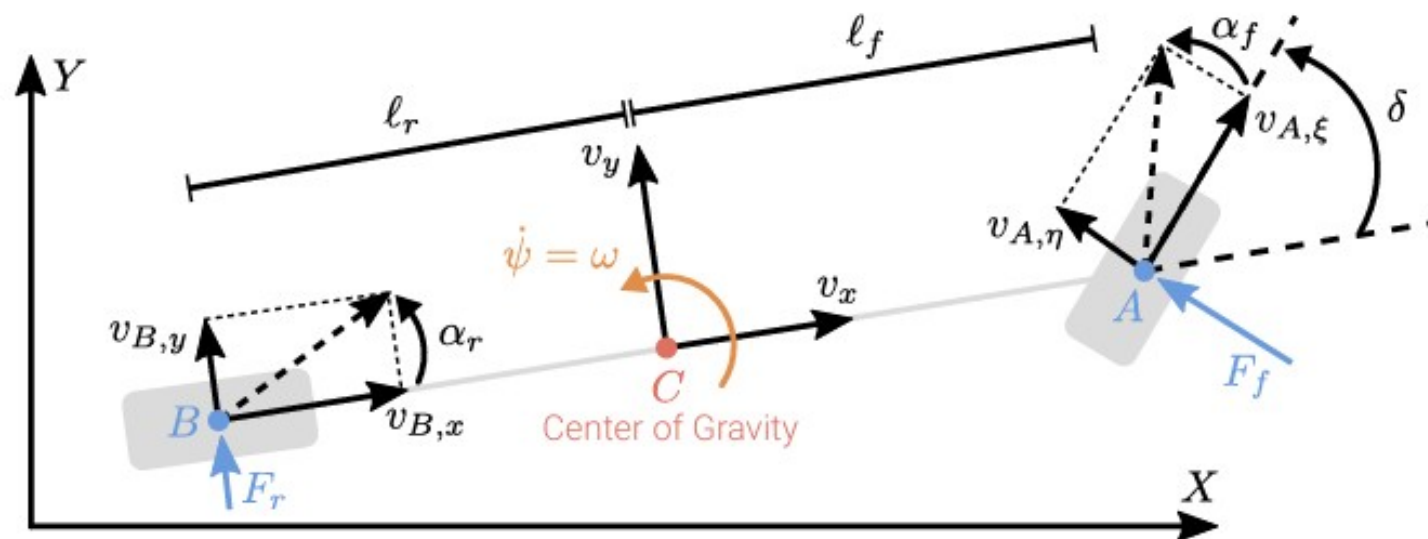
$$v_{B,x} = v_x \quad v_{B,y} = v_y - \omega l_r$$

$$v_{A,x} = v_x \quad v_{A,y} = v_y + \omega l_f$$

$$v_{A,\xi} = v_{A,x} \underbrace{\cos(\delta)}_{\approx 1} + v_{A,y} \underbrace{\sin(\delta)}_{\approx \delta}$$

$$v_{A,\eta} = -v_{A,x} \underbrace{\sin(\delta)}_{\approx \delta} + v_{A,y} \underbrace{\cos(\delta)}_{\approx 1}$$

Dynamic Bicycle Model



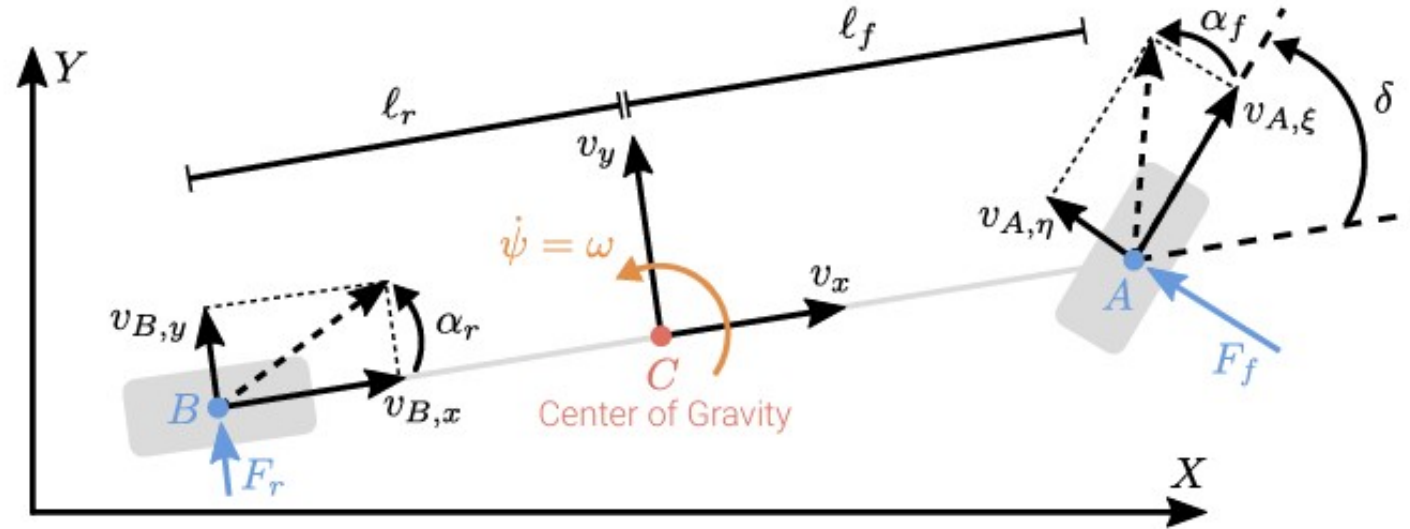
Tire Forces:

$$F_r = -c_r \frac{v_{B,y}}{v_{B,x}} = -c_r \frac{v_y - \omega l_r}{v_x}$$

$$F_f = -c_f \frac{v_{A,\eta}}{v_{A,\xi}} = -c_f \frac{-v_x \delta + v_y + \omega l_f}{v_x + (v_y + \omega l_f) \delta} \approx c_f \delta - c_f \frac{v_y + \omega l_f}{v_x}$$

Last approximation due to: $v_x \gg (v_y + \omega l_f) \delta$

Dynamic Bicycle Model

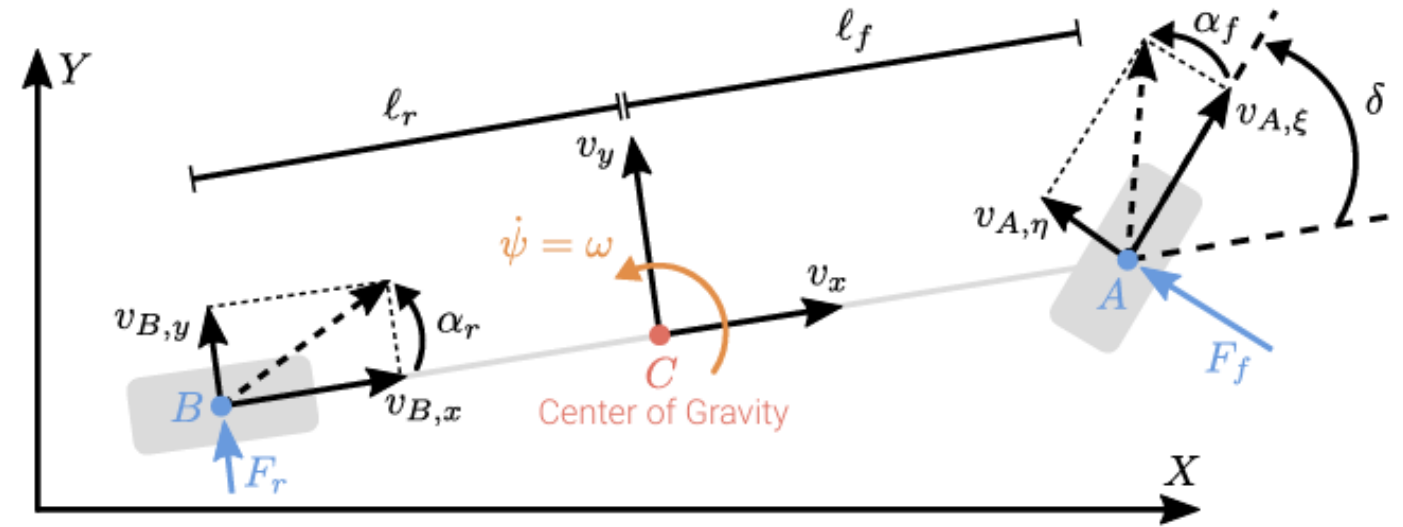


State Space Representation:

$$m(\dot{v}_y + \omega v_x) = \underbrace{-c_r \frac{v_y - \omega l_r}{v_x}}_{=F_r} + \underbrace{c_f \delta - c_f \frac{v_y + \omega l_f}{v_x}}_{=F_f}$$

$$I_z \dot{\omega} = -l_r \underbrace{\left(-c_r \frac{v_y - \omega l_r}{v_x} \right)}_{F_r} + l_f \underbrace{\left(c_f \delta - c_f \frac{v_y + \omega l_f}{v_x} \right)}_{=F_f}$$

Dynamic Bicycle Model



State Space Representation:

$$\begin{bmatrix} \dot{v}_y \\ \dot{\psi} \\ \dot{\omega} \end{bmatrix} = \begin{bmatrix} -\frac{c_r + c_f}{mv_x} & 0 & \frac{c_r l_r - c_f l_f}{mv_x} - v_x \\ 0 & 0 & 1 \\ \frac{l_r c_r - l_f c_f}{I_z v_x} & 0 & -\frac{l_f^2 c_f + l_r^2 c_r}{I_z v_x} \end{bmatrix} \underbrace{\begin{bmatrix} v_y \\ \psi \\ \omega \end{bmatrix}}_{\text{State}} + \underbrace{\begin{bmatrix} \frac{c_f}{m} \\ 0 \\ \frac{c_f}{I_z} l_f \end{bmatrix} \delta}_{\text{Input}}$$

Can be augmented by the global position to a nonlinear state space model

Summary

- ▶ A vehicle can be modeled as a rigid body
- ▶ It is subject to holonomic and non-holonomic constraints
- ▶ The bicycle model approximates the vehicle using 2 wheels
- ▶ The kinematic bicycle model assumes no wheel slip (low speeds)
- ▶ However, modeling tires requires to consider slip
- ▶ Sliding friction is smaller than static friction
- ▶ We want to operate in the static friction area of the force curve
- ▶ The circle of forces tells us that lat. and long. forces are dependent
- ▶ The dynamic bicycle model takes into account tire forces and wheel slip